

Problem statement:

- Background
- The Pradhan Mantri Anusuchit Jaati Abhyuday Yojana (PM-AJAY) aims to reduce poverty among Scheduled Caste (SC) communities through livelihood promotion, skill development, and enterprise support under its Grant-in-Aid (GIA) component. A major challenge in implementation is the identification of appropriate skill training pathways that align with both the aspirations of beneficiaries and the actual livelihood opportunities available in their local regions.
- Many target beneficiaries face barriers such as low digital literacy, limited awareness of modern trades, language constraints, and difficulty navigating text-heavy digital systems. As a result, there is often a mismatch between enrolled training programs and the beneficiary's interests, capabilities, or local market demand, leading to high dropout rates and poor post-training employment outcomes.
- To improve inclusion and effectiveness, there is a need for an AI-enabled conversational system that can interact naturally in regional languages and dialects, understand beneficiary aspirations, assess skill gaps, and recommend suitable NSQF aligned livelihood opportunities in and around the beneficiary.
- Basic Issues under GIA Component:
- Lack of proper road map and Planning of the Perspective plans from execution to implementation
- Identification of the participants Trained and skilled Financial consultants
- Job placement issue after the skilling programme
- Coordination Issues among the corporation, Ministry/Departments
- Inadequate Technical and support team at ground level
- Detailed Description The proposed solution should be an AI-driven, multilingual, voice-based virtual livelihood assistant capable of conducting conversational interviews with beneficiaries from aspirational SC communities. Instead of relying on traditional form-filling methods, the system should use voice interactions to collect information such as:
- Educational background
- Existing or traditional family occupations
- Current livelihood activities
- Skills and interests
- Mobility and physical constraints
- Preference for self-employment or wage employment
- Local economic realities and opportunities The assistant should support regional languages and dialects to ensure accessibility for users with low literacy or limited digital exposure. The interaction should feel empathetic and conversational rather than administrative. The collected information should be analyzed using AI/MLbased profiling and recommendation mechanisms to identify:
- Suitable NSQF-aligned training programs
- Relevant trades and livelihood pathways
- Skill gaps requiring intervention
- Region-specific employment or enterprise opportunities The system should also function effectively in lowconnectivity and low-tech environments through deployment channels such as:
- IVR-based phone calls for feature phone users

- WhatsApp voice-note interfaces Lightweight mobile or kiosk-based solutions
- Expected Solution:

An AI-powered multilingual voice assistant application designed to help SC beneficiaries under PM-AJAY identify suitable skill training and livelihood opportunities. The app will support regional languages and local dialects, allowing users to interact through simple voice conversations instead of text-based forms.

PROBLEM STATEMENT DETAILS

Problem Statement ID	26097
Problem Statement Title	AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY
Description	- Background - The Pradhan Mantri Anusuchit Jaati Abhyuday Yojana (PM-AJAY) aims to reduce poverty among Scheduled Caste (SC) communities through livelihood promotion, skill development, and enterprise support under its Grant-in-Aid (GIA) component. A major challenge in implementation is the identification of appropriate skill training pathways that align with both the aspirations of beneficiaries and the actual livelihood opportunities available in their local regions. - Many target beneficiaries face barriers such as low digital literacy, limited awareness of modern trades, language constraints, and difficulty navigating text-heavy digital systems. As a result, there is often a mismatch between enrolled training programs and the beneficiaryâ€™s interests, capabilities, or local market demand, leading to high dropout rates and poor post-training employment outcomes. - To improve inclusion and effectiveness, there is a need for an AI-enabled conversational system that can interact naturally in
Organization	Ministry of Social Justice and Empowerment (MoSJE)
Department	Department of Social Justice and Empowerment
Category	Software
Theme	Agriculture, FoodTech & Rural Development

Research:::

An AI-driven, multilingual, voice-based livelihood assistant is feasible and well-aligned with PM-AJAY’s GIA objectives, especially if it combines conversational data collection, NSQF-aware profiling, and region-specific opportunity mapping, and is delivered via IVR, WhatsApp, and lightweight apps for low-literacy SC beneficiaries. A carefully phased, hybrid (cloud + edge) architecture with open-source speech/LLM models, integrated with NSQF and Skill India infrastructure, can directly address planning gaps, misaligned skilling, and post-training placement issues under the scheme.[\[ncvet.gov\]](#)[\[pib.gov\]](#)[\[socialjustice.gov\]](#)[\[pib.gov\]](#)[\[insightsonindia\]](#)

Policy and scheme context

- PM-AJAY is a centrally sponsored scheme of the Ministry of Social Justice & Empowerment (MoSJE) for integrated socio-economic development of Scheduled Castes, including livelihood promotion and skill development.[\[socialjustice.gov\]](https://socialjustice.gov)[\[dosje.gov\]](https://dosje.gov)[\[pib.gov\]](https://pib.gov)
 - Under the Grants-in-Aid (GIA) component, states receive funds for district/state-level projects such as skill development, comprehensive livelihood projects, and income-generation assets for SC households.[\[insightsonindia\]](https://insightsonindia)[\[pib.gov\]](https://pib.gov)[\[socialjustice.gov\]](https://socialjustice.gov)
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Why current GIA implementation struggles

- Many states submit Annual Action Plans and perspective plans, but field-level implementation suffers from poor planning, weak MIS usage, and limited real-time linkage between beneficiary profiling, training enrolment, and post-training tracking.[\[socialjustice.gov\]](https://socialjustice.gov)[\[insightsonindia\]](https://insightsonindia)[\[pib.gov\]](https://pib.gov)
- NSQF has thousands of aligned qualifications (over 9,500 approved, with 3,122 active as of December 2025), yet field-level staff and beneficiaries often cannot navigate this complexity, leading to ad-hoc or supply-driven training choices.[\[pib.gov\]](https://pib.gov)[\[skills.gujarat.gov\]](https://skills.gujarat.gov)[\[dvet.gov\]](https://dvet.gov)

Your problem statement correctly highlights additional implementation issues (these align with broader skilling ecosystem challenges in India):

- Lack of perspective planning and roadmaps for GIA-funded livelihood projects.
 - Difficulty in identifying appropriate beneficiaries and skilled resource persons (trainers, financial counsellors).
 - Weak job placement linkages and post-training handholding.
 - Poor coordination between corporations, departments, and ministries.
 - Inadequate technical support teams at block/village level.
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Beneficiary-level barriers (ground realities)

From national skilling experience and initiatives like NSQF and Skill India, typical barriers for SC beneficiaries include:[\[ncvet.gov\]](https://ncvet.gov)[\[pib.gov\]](https://pib.gov)[\[nielit\]](https://nielit)

- Low or uneven literacy and digital literacy, making text-heavy portals and forms inaccessible.
- Limited awareness of modern trades, NSQF levels, and sector skill council offerings; many only know traditional occupations or casual work options.
- Language and dialect constraints; comfort mainly in local languages, mixed registers, or dialects (e.g., Bundeli, Bhojpuri, Marwari, etc.).

- Complex forms and multiple agencies (training centres, banks, line departments), causing fatigue and dropout at the application stage.

This leads to:

- Mismatch between beneficiary aspirations and enrolled courses (e.g., enrolling in a trade only because it's available locally, not because it fits their profile).
 - High dropout rates due to wrong trade selection, distance, family constraints, or unrealistic expectations.
 - Poor placement and livelihood outcomes, which in turn reflect as underutilization of GIA funds or low impact.
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NSQF and why alignment matters

- NSQF is India's competency-based framework organizing qualifications into levels based on knowledge, skills, and responsibility; each level captures complexity and autonomy of work. [ncvet.gov][pib.gov][[nielit](https://nielit.gov.in/)]
- Qualifications are defined through sector skill councils (SSCs) as Qualification Packs with National Occupational Standards (NOS), which describe specific job roles and competencies. [dvet.gov][[nielit](https://nielit.gov.in/)][[hyring](https://hyring.org/)]

For PM-AJAY, NSQF alignment matters because:

- It creates a standardized structure for mapping beneficiaries to appropriate levels (e.g., Level 2–4 for basic trades, Level 5+ for supervisory/advanced roles). [[hyring](https://hyring.org/)][iitms.co][ncvet.gov]
 - It supports mobility: a learner can progress from a basic livelihood skill (Level 1–2) to more advanced roles with time.
 - Many government skilling and placement systems (Skill India, NSDC, SIDH, etc.) already use NSQF, so integration avoids duplication. [pib.gov][pib.gov]
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Existing precedents for AI/voice assistants in skilling

There are real-world examples in India showing feasibility of AI-driven, multilingual, voice-based assistants:

- The Skill India Assistant (SIA) chatbot launched by the Ministry of Skill Development and Entrepreneurship (MSDE) uses an AI model (based on Meta's LLaMA) to help users discover courses, locate centres, and access job listings via WhatsApp, with multilingual and voice support. [pib.gov]

- Voice-first, offline-capable applications like the “Vyapar AI” business assistant demonstrate that Hindi/Hinglish voice interfaces can be built for low-resource small enterprises, with offline speech-to-text and WhatsApp integration.[\[ijert\]](#)

These show:

- WhatsApp and low-bandwidth channels can carry conversational AI for skilling and livelihoods.
 - On-device or offline-first speech pipelines are feasible for Indian languages, if models are optimized.[\[pib.gov\]](#)[\[ijert\]](#)
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Problem reframing for your solution

Reframed in hackathon-style language, the core problem is:

How can we design an AI-driven, multilingual, voice-first virtual assistant that (a) understands SC beneficiaries’ aspirations and constraints, (b) maps them to NSQF-aligned skilling and livelihood pathways rooted in local opportunities, and (c) operates over low-tech channels like IVR and WhatsApp to improve outcomes under the PM-AJAY GIA component?

Sub-problems your solution must address:

1. Accessible beneficiary profiling without forms (voice-only, in local languages).
 2. Intelligent matching of beneficiary profile to NSQF roles and trades.
 3. Integration with local opportunity data (jobs, enterprises, schemes, markets).
 4. Low-bandwidth, inclusive delivery channels (feature phone + basic smartphone).
 5. Monitoring: converting conversations into actionable MIS data for MoSJE and implementing agencies.
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High-level solution vision

Solution name idea: “JeevanMitra – PM-AJAY Livelihood Assistant” (you can rename as per team preference).

Core idea:

- A multilingual, voice-first virtual livelihood assistant that talks to beneficiaries in their preferred language/dialect, asks empathetic questions, builds a profile, identifies skill gaps, and recommends NSQF-aligned training and livelihood options available in their region.

- Delivered via IVR calls, WhatsApp voice notes, and lightweight Android app/kiosk, with offline-friendly components for low-connectivity blocks.

Key outcome metrics:

- Higher alignment between beneficiary interest and course selection.
 - Reduced dropout during and after training.
 - Improved tracking of placements and self-employment outcomes under PM-AJAY GIA.
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User journeys (end-to-end)

1. Feature-phone user (IVR journey)

- Beneficiary receives a call/SMS from district SC development corporation or dials a toll-free number.
- IVR greets in local language, offers language selection, and uses short, conversational prompts instead of forms.
- The system collects: education, family occupation, current work, constraints (mobility, childcare, disability), aspirations, risk appetite, and preference for self or wage employment—entirely via voice.
- AI extracts structured fields + narrative text from speech, creates a profile, and runs a recommendation engine.
- Towards the end of the call, the assistant explains 1–3 suitable options in simple language and optionally sends a follow-up SMS with contact details of local training partners or schemes.

2. Smartphone user (WhatsApp + app journey)

- Beneficiary opens a WhatsApp number or kiosk app in a CSC/PM-AJAY centre; they hold down mic and speak.
 - The assistant chats via voice/short text, repeating the same profiling, but can also show simple visual cards (e.g., “Electrician Level 3 – 4 months, nearby centre 8 km away”).
 - Recommendations are saved to a profile; the user can click to:
 - Apply or pre-register to a training program.
 - Request a callback from a human counsellor.
 - See small explainers (audio) about each trade.
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AI/ML core components

You can structure the solution around these components (good for your architecture diagram / documentation):

1. Speech interface layer

- ASR (automatic speech recognition) models for Hindi + major regional languages, plus handling of code-mixed speech (Hinglish).
- Text-to-speech (TTS) voices tuned for friendly, empathetic tone, not robotic, with male/female voice options.

2. Natural language understanding (NLU)

- Intent classification (e.g., “interested in agriculture”, “wants to migrate for work”, “prefers home-based self-employment”).
- Entity extraction for education level, skills, constraints, experience, location, etc.

3. Beneficiary profiling engine

- Converts conversational responses into a structured “profile vector” with dimensions like: education level, NSQF readiness level, sector interest, mobility, gender, age, prior skills, and risk preference.
- Uses rules + ML models to assign a tentative NSQF entry level and possible progression pathways based on NSQF definitions. [\[ncvet.gov\]](https://ncvet.gov) [\[pib.gov\]](https://pib.gov) [\[hyring\]](https://hyring)

4. Recommendation engine (NSQF + local opportunities)

- Knowledge graph of NSQF qualifications, job roles, and sectors.
- Integration (where possible) with Skill India, NSDC/SIDH APIs, and state skill missions to pull course availability and training centre data. [\[pib.gov\]](https://pib.gov) [\[pib.gov\]](https://pib.gov)
- Mapping of regional opportunities: local employer demands, agriculture value-chain opportunities, traditional crafts, SHG/producer company initiatives, etc.

5. Policy + GIA integration layer (MIS)

- Converts each interaction into:
 - Candidate record with profile and recommended trades.
 - Suggested interventions (training, financial literacy, enterprise support).
- Pushes this data to a PM-AJAY GIA MIS dashboard for district/state officials to plan batches, trainers, and financial linkages.

Addressing the “basic issues” one-by-one

1. Lack of roadmap and perspective planning

- Aggregate conversational data (sector interest, preferred locations, constraints) to generate demand heatmaps for each district/block.
- Use these for:
 - Evidence-based Annual Action Plans (AAPs) under PM-AJAY (e.g., “150 women in Block X want home-based food processing Level 3 courses”).
 - Planning of trainers, centres, and supporting infrastructure under GIA.

2. Identification of participants and skilled resource persons

- The assistant acts as a scalable “front-door”: any potential beneficiary can call/chat, eliminating dependence on physical camps alone.
- Profiles tagged as “high risk/high motivation” can be flagged to field staff or NGOs for targeted counselling or home visits.
- The system can also be extended to register local resource persons (e.g., retired ITI instructors, successful entrepreneurs) through separate flows, building a pool of mentors/consultants.

3. Job placement after skilling

- Integrate with job listing feeds (from Skill India, state employment portals, local industry associations) and match trained candidates based on skills, location, and preferences. [pib.gov][pib.gov]
- For self-employment, the assistant can:
 - Provide audio guides on how to register enterprises, access credit schemes, or join SHGs.
 - Trigger “leads” to District Industries Centres, banks, or NRLM/State Rural Livelihood Missions.

4. Coordination across corporations and departments

- Use the assistant-generated MIS as a single source of truth for:
 - Demand-side (beneficiary) data.
 - Supply-side data (courses, schemes, infrastructure).
- Role-based dashboards (state corporations, MoSJE, skill departments) can view and act on this data instead of siloed spreadsheets.

5. Inadequate technical support at ground level

- Because the assistant handles routine profiling and basic counselling, field staff focus on complex cases, mobilization, and community trust-building.
- Kiosk deployments (CSC centres, PM-AJAY model villages, panchayat offices) allow even tech-shy beneficiaries to interact with AI with assistance from a local operator.

Operating in low-connectivity and low-tech environments

Given many SC-dominated villages have patchy connectivity, you should explicitly design for:

- **IVR-first design:** Core flows must work via voice-only calls, using DTMF fallback for menu selection.

- **Offline/edge models:** Use compressed ASR and NLU models on local servers at district centres or in offline-first Android apps (similar to approaches used in other offline-capable voice assistants).[\[ijert\]](#)
 - **Low-bandwidth WhatsApp:** Following SIA's approach, rely on WhatsApp for text + audio for smartphone users, as it is already widely used and works under low bandwidth.[\[pib.gov\]](#)
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Data, privacy, and ethics considerations

For a government-facing hackathon solution, highlight:

- Explicit consent prompts (audio) before data collection and clear explanation of how data will be used.
 - Data minimization—only collect necessary data (no unnecessary sensitive questions).
 - Anonymized/aggregated analytics for planning; personally identifiable data accessible only to authorized staff.
 - Bias mitigation: ensure recommendations are not reinforcing stereotypes (e.g., women only being shown certain trades).
 - Support for persons with disabilities (e.g., audio-only flows for visually impaired, slow speech modes, repeat options).
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Implementation roadmap (phased)

Phase 1 – Prototype (Hackathon-ready)

- Scope: One language (Hindi) + one pilot district, subset of NSQF trades (e.g., construction, electrical, tailoring, agriculture processing).
- Deliverables:
 - Simple IVR or Twilio-like call interface + WhatsApp bot.
 - Rule-based profiling + basic recommendation logic using predefined mappings.
 - Admin dashboard (React + backend) to see profiles and recommendations.

Phase 2 – Pilot with state/district

- Add more languages/dialects and more trades.
- Integrate with existing skill portals (Skill India Digital Hub, state NSQF portals) where APIs are feasible.[\[pib.gov\]](#)[\[pib.gov\]](#)
- Collect feedback from field staff, NGOs, and beneficiaries to refine questions and scripts.

Phase 3 – Scale-up and deep AI

- Introduce ML-based recommendation models (e.g., collaborative filtering or supervised learning based on outcome data).
 - Add advanced features like:
 - Personal progression plans (Level 2 → Level 3 → Level 4).
 - Early dropout prediction based on conversation signals.
 - Auto-summarized case notes for human counsellors.
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Indicative tech stack (for your project plan)

You can propose something like this (adjust based on tools you like):

- **Frontend / Channels:**
 - WhatsApp (Cloud API or BSP integration).
 - IVR (telephony providers or open-source tools like Asterisk/FreeSWITCH).
 - Web/Kiosk: React, Tailwind CSS (your strength), deployed on Vercel or similar.
 - **Backend:**
 - Python/Node microservices for ASR, NLU, recommendation engine.
 - PostgreSQL / MongoDB for storing profiles and interactions.
 - **AI models:**
 - Open-source ASR for Indian languages (e.g., Indic-focused ASR models) deployed as Docker containers.
 - LLM/NLU: open-source LLaMA or other instruction models fine-tuned for short-turn conversations in Hindi + regional languages (inspired by SIA using LLaMA). [[pib.gov](https://pib.gov.in)]
 - Lightweight on-device models where offline is essential (like in Vyapar AI-style solutions). [[ijert](https://ijert.org)]
 - **Integration:**
 - APIs/web-scrapers for NSQF qualification data and Skill India course directories. [[ncvet.gov](https://ncvet.gov.in)][[pib.gov](https://pib.gov.in)][[hyring](https://hyring.in)]
 - Optional integration with PM-AJAY portal MIS if APIs are provided. [[socialjustice.gov](https://socialjustice.gov.in)]
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How you can structure your hackathon/documentation write-up

Here is a structure you can directly adapt for your project report/pitch:

1. **Introduction and policy context**
 - Brief on PM-AJAY, GIA, and NSQF; highlight PM-AJAY objective of socio-economic upliftment of SCs through livelihood support and skill development. [socialjustice.gov][[insightsonindia](https://insightsonindia.com)][pib.gov]
 2. **Problem statement**
 - Misalignment between beneficiary aspirations and training choices, high dropout, poor placement.
 - Barriers: low literacy, digital divide, language gaps, fragmented planning.
 3. **Objectives of the proposed system**
 - Make beneficiary onboarding conversational and voice-first.
 - Provide NSQF-aligned, locally relevant training and livelihood recommendations.
 - Generate real-time data to improve PM-AJAY GIA planning and monitoring.
 4. **System design and architecture**
 - Diagram showing: Channels → ASR/TTS → NLU → Profiling Engine → Recommendation Engine → MIS Dashboard.
 - Highlight low-connectivity design features, multilingual support, and integration points.
 5. **AI/ML methodology**
 - Techniques for profiling, NSQF mapping, and recommendation.
 - Future scope: model training from outcome data, personalization, and explainable recommendations.
 6. **Implementation plan**
 - Phase-wise rollout, pilot districts, language coverage.
 7. **Impact assessment**
 - KPIs: number of profiled beneficiaries, training alignment score, dropout reduction, % placed/self-employed, GIA fund utilization.
 8. **Risks and mitigations**
 - Data privacy, algorithmic biases, technical failure in low network areas, adoption challenges among field staff.
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system design/workflow/sreenlayouts of ui ux/backend archecture :

PM-AJAY voice livelihood assistant: design blueprint

Build this as a voice-first guidance system, not just a chatbot. A beneficiary speaks about their interests and circumstances; the system turns that conversation into a reviewable profile, matches it against verified NSQF qualifications and locally confirmed opportunities, and helps a field worker follow through. This fits PM-AJAY's Grants-in-Aid focus on skill development and comprehensive livelihood projects. [socialjustice.gov]

For a hackathon, I recommend one district, Hindi plus one locally relevant language or dialect, a small verified qualification catalogue, and a working field-worker dashboard. Treat IVR and WhatsApp as later channels unless your team can integrate and test them properly.

1. System design

BENEFICIARY CHANNELS

Mobile web app / assisted kiosk

Later: IVR phone call / WhatsApp voice note



VOICE INTERACTION

Record → speech-to-text → conversational interview

→ confirm answers aloud → text-to-speech response



PROFILE AND MATCHING

Structured beneficiary profile

+ verified NSQF qualifications

+ locally verified training centres/opportunities

→ eligibility filters → ranked recommendations



ACTION AND FOLLOW-UP

Beneficiary hears top 3 options

Field worker reviews, corrects and assists

Training referral → enrolment → completion → livelihood outcome



DISTRICT DASHBOARD

**Demand by trade, referral status, drop-off points,
training capacity and placement follow-up**

The important separation is qualification versus availability. An NSQF-aligned qualification describes a recognized learning pathway; it does *not* prove that a nearby centre has an open batch or that a job exists. Use the National Qualifications Register (NQR) to verify qualification details, then maintain a separate, dated local dataset for centres, seats, employers and enterprise opportunities. NQR is the official public record for NSQF-aligned qualifications. [ncvet.gov][nqr.gov]

Component	Responsibility	Hackathon implementation
Beneficiary app	Language selection, voice interview, spoken results	Next.js mobile-first PWA
Voice layer	Transcription and spoken replies	Replaceable STT/TTS provider adapters
Interview engine	Ask one question at a time; extract and confirm answers	LLM with structured JSON output
Matching engine	Filter and rank verified pathways	Deterministic rules and weighted scores
Data layer	Profiles, qualifications, local availability, referrals	Supabase Postgres and Storage
Field-worker portal	Correct profiles, verify opportunities, manage referrals	Role-based dashboard
Reporting	Track implementation and outcomes	Aggregated district views

Do not let the LLM invent courses, centres, seats, salaries or jobs. Give it approved catalogue records and let it explain the recommendations in the beneficiary's language.

2. User and operational workflow

1. **Welcome and consent:** Play a short explanation in the selected language: what information will be collected, why, and whether voice recordings will be kept. Provide a clear way to stop or request help. Design the notice and affirmative consent flow around applicable Indian data-protection requirements; the DPDP Act sets out requirements concerning notice, consent, safeguards and correction or erasure rights, subject to applicable provisions and rules.[meity.gov]
2. **Conversational interview:** Ask about district/block, education, existing work, interests, prior skills, training availability, travel radius, accessibility needs and preference for wage work or self-employment. Allow “skip” and “I don’t know.”
3. **Confirm the profile:** Read back important answers: “You said you can travel up to 10 km and prefer self-employment. Is that right?” Correct transcription errors before matching.
4. **Create a shortlist:** Check entry requirements and constraints first. Rank remaining qualifications using interests, existing skills, travel feasibility and evidence of local demand. Return up to three distinct options.
5. **Explain and act:** For each option, say *why it fits*, the likely skill gap, the verified qualification details, and the status of any nearby training opportunity. Offer “listen again,” “compare,” or “request a field-worker call.”
6. **Human review and follow-up:** A field worker verifies uncertain facts, contacts the person with permission, records referral and enrolment status, and follows up on completion and livelihood outcomes.
7. **Planning loop:** District staff see aggregated interest versus training capacity and referral bottlenecks. This makes the system useful for perspective planning, coordination and post-training tracking—not just initial recommendations.

Matching logic: Start with hard filters such as qualification entry requirements and a beneficiary’s stated constraints. Then use an explainable score—for example, interest 30%, prior skills 20%, feasible access 20%, locally verified demand 20%, and stated work preference 10%. These are *proposed product weights*, not government rules; test them with beneficiaries and field workers. If local availability is unverified, show “qualification match—local batch not confirmed”, not “apply now.”

An illustrative result might say: “Your experience helping with a family food business makes *Small Food Business Operator* worth exploring. It appears in NQR as a Level 2, 300-hour qualification. A worker still needs to confirm whether a suitable batch is available near you.” The qualification details are verifiable; the nearby batch in this example is deliberately not asserted.[nqr.gov]

3. Screen layouts and UX

Use large controls, short spoken prompts, high-contrast text, visible progress and no mandatory typing. Keep the interface usable by a beneficiary *or* a worker operating the phone on their behalf.

Screen 1 — Welcome and language

आपकी आजीविका, आपकी पसंद	
Find training and work pathways	
Choose a language / भाषा चुनें	
[हिन्दी ▼]	
[🔊 Hear how this works]	
[🎤 Start speaking]	
Assisted by a field worker?	
[Switch to assisted mode]	

Screen 2 — Consent

Before we begin	
[🔊 Play short explanation]	
We use your answers to suggest	
training and livelihood paths.	
You can skip questions or stop.	

Voice recording: [Do not keep ▼]
[I agree and continue]
[Ask a worker for help]

Screen 3 — Voice interview

Question 3 of about 8

● ● ● ○ ○ ○ ○ ○

“What work do you enjoy doing?”

[ Replay question]

[ Hold to talk]

Heard: “I like repairing...”

[✓ Correct] [↺ Say again]

[Skip] [End for now]

Screen 4 — Confirm profile

Did we understand correctly?

District: [Example district ]

Education: [Class 10 ]

Interest: [Repair work ]

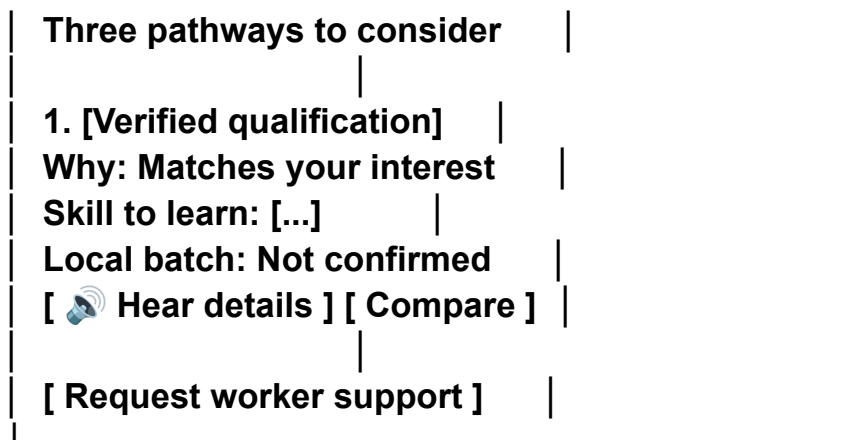
Travel: [Up to 10 km ]

Preference: [Self-employment ]

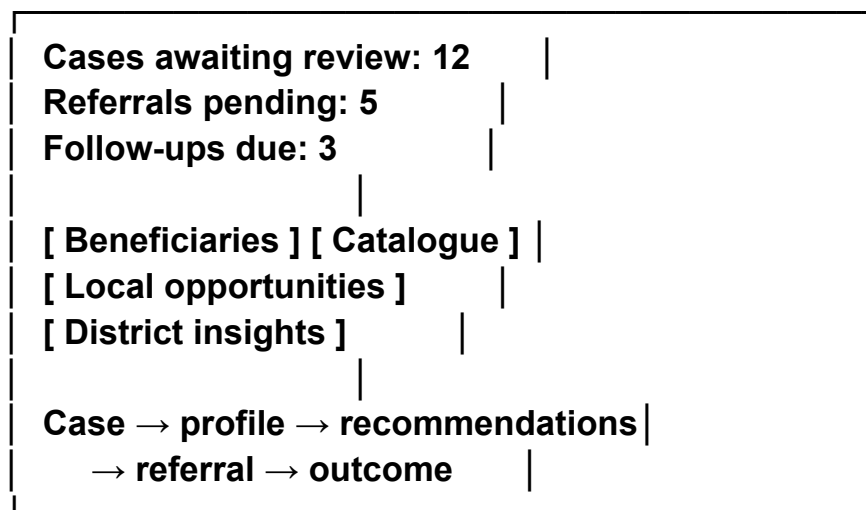
[ Read my answers]

[Find my options]

Screen 5 — Recommendations



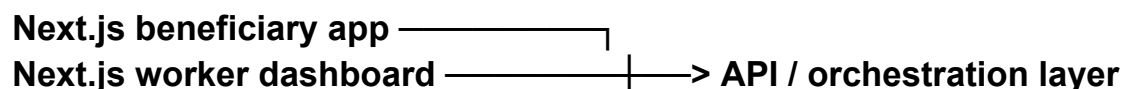
Screen 6 — Field-worker dashboard

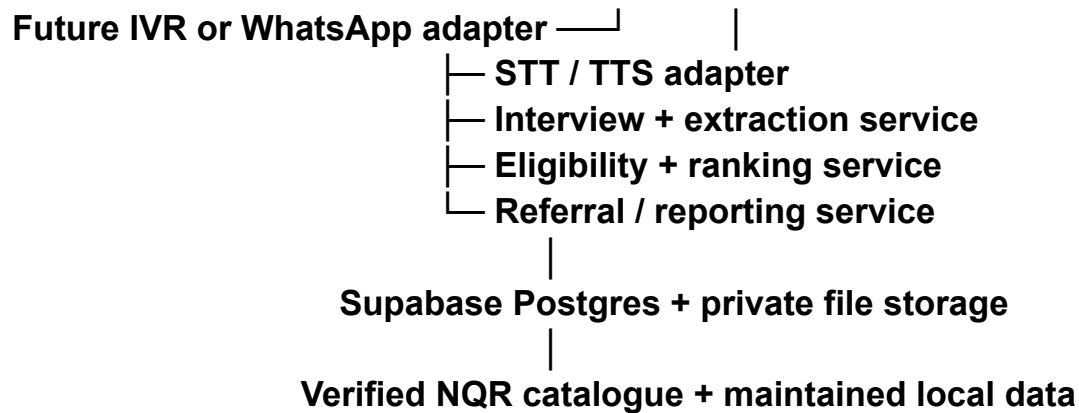


The counts above are wireframe placeholders, not real programme data. In production, keep sensitive profile details out of broad district dashboards.

4. Backend architecture

Suggested stack for your workflow: Next.js + TypeScript for the app and dashboard, Supabase for Postgres/Auth/Storage, and server-side functions for AI orchestration. Use one voice-provider interface so you can change speech services without rewriting the interview. Keep API keys server-side.





Core tables

Table	Essential fields
beneficiaries	id , preferred language, district/block, contact preference, created date
consents	beneficiary_id , purpose, notice version, choice, timestamp
interview_sessions	id , beneficiary, channel, status, last question, language
profile_answers	beneficiary_id , field, value, confirmation status, source
qualifications	NQR identifier/link, title, sector, NSQF level, entry criteria, skills, verification date
local_opportunities	qualification ID, centre/employer, location, batch or vacancy status, source, verified date
recommendations	beneficiary, qualification, score, explanation, data snapshot
referrals	beneficiary, opportunity, assigned worker, status, next follow-up

outcomes	enrolment, completion, employment/self-employment status, follow-up date
audit_events	actor, action, record, timestamp

Apply row-level permissions: beneficiaries see their own information; assigned workers see only their cases; district admins see appropriately scoped records and aggregated reports. Store audio privately only when needed, set a retention policy, and avoid collecting caste documents or precise location unless a clearly defined workflow requires them. Keep programme eligibility verification separate from AI-generated career guidance.

Indicative API endpoints: **POST /api/interview/start**, **POST /api/interview/turn**, **POST /api/profile/confirm**, **GET /api/recommendations/:beneficiaryId**, **POST /api/referrals**, and worker-only catalogue verification endpoints. Each interview turn should return a spoken reply, a display transcript, extracted fields, confidence flags and the next question. Validate the structured output on the server before saving it.

For low connectivity, cache the interface and allow an assisted kiosk to queue *consented* answers for later sync. Do not promise fully offline cloud AI. IVR needs telephony integration; WhatsApp voice notes need a messaging-provider integration. Both should feed the same interview and matching backend, but each needs its own consent, identity and error-handling flow.

5. Build plan and demo

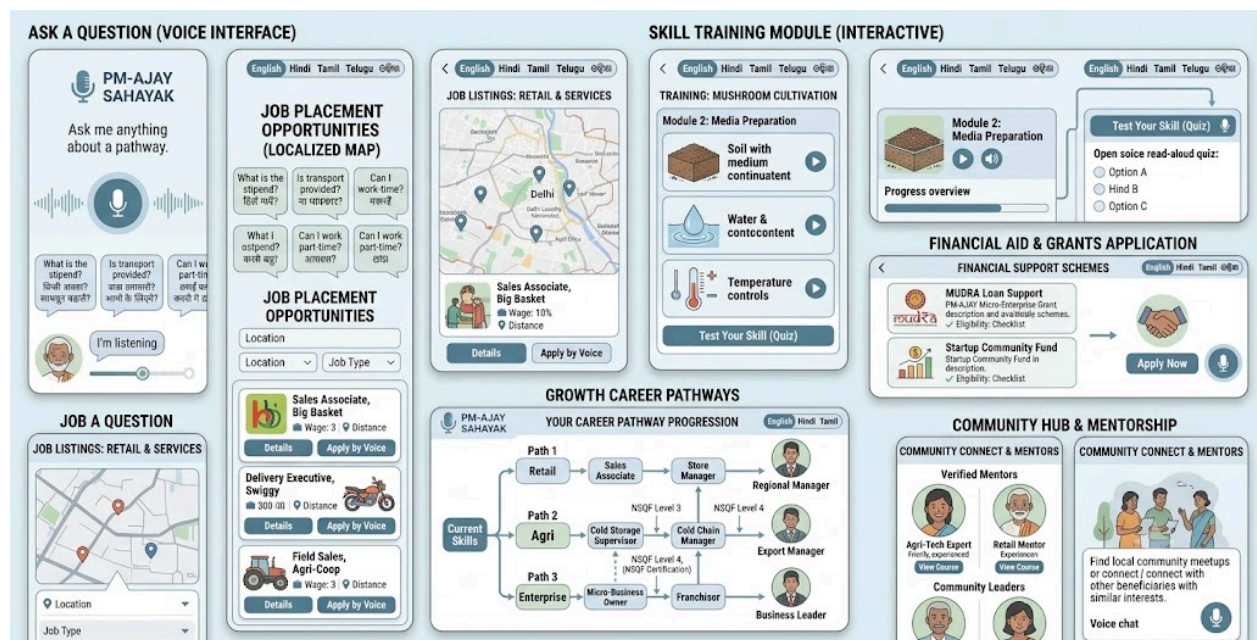
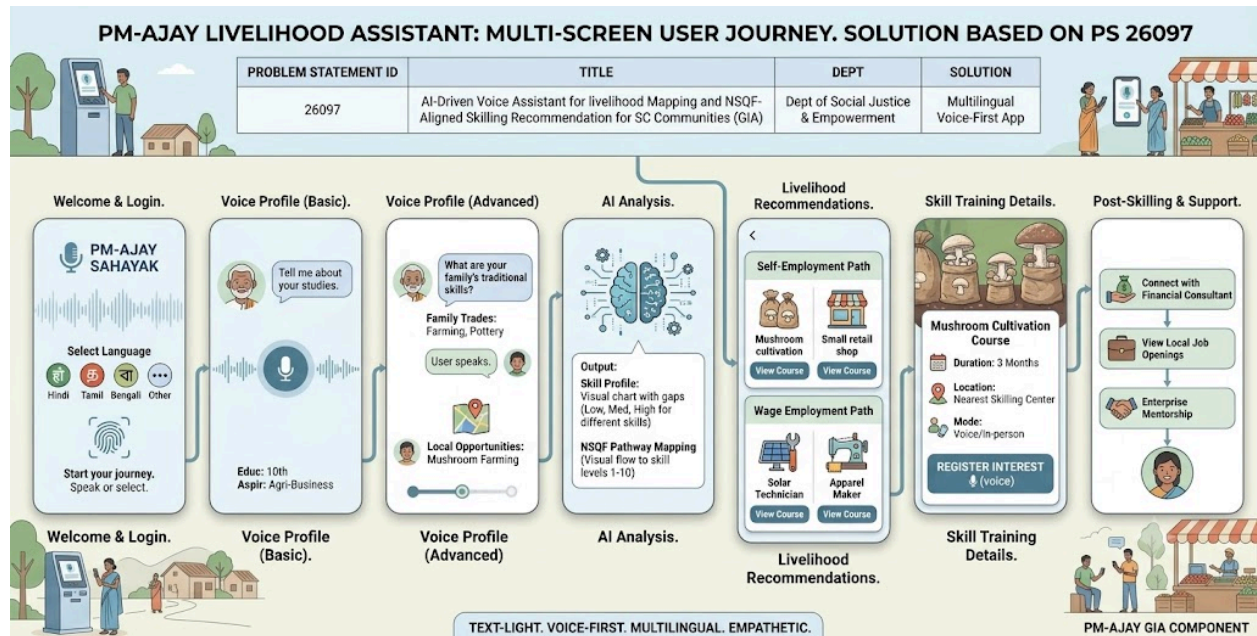
Phase	Build	Proof it works
1. Data foundation	Import a small, manually verified NQR sample; create separate sample local opportunities	Every result shows its source and verification status

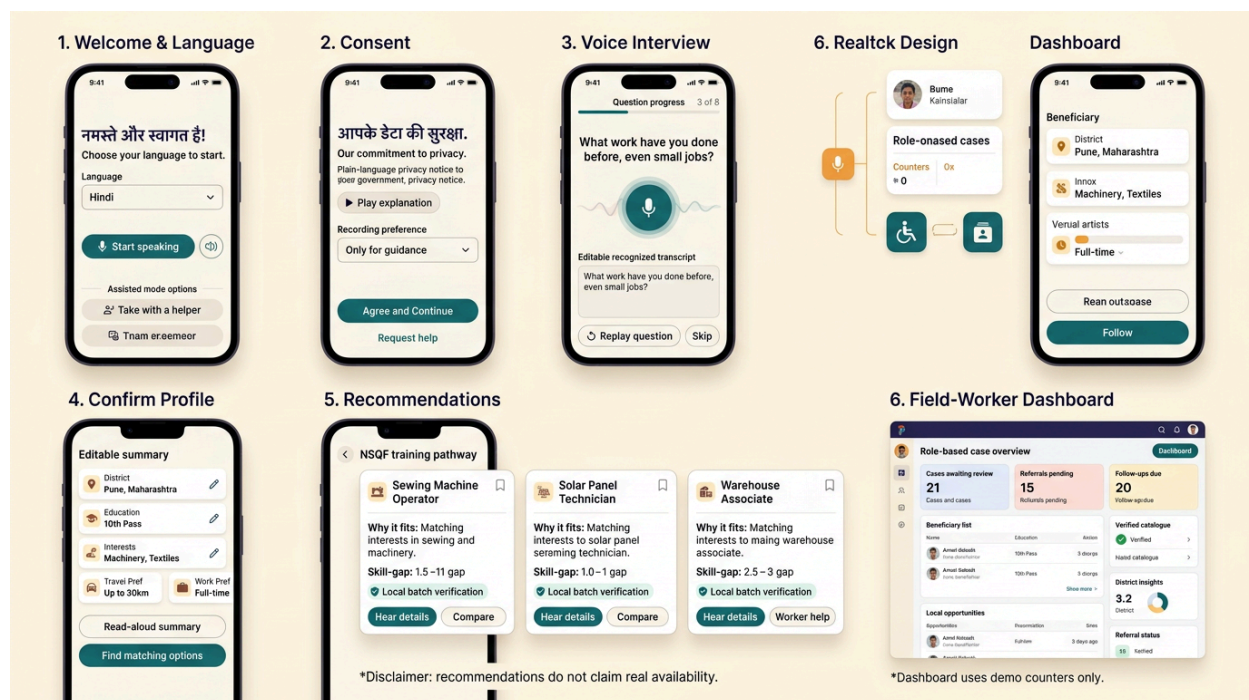
2. Beneficiary journey	Language choice, consent, voice interview, answer correction	A user completes the journey without typing
3. Recommendations	Eligibility checks, top-three ranking, spoken explanations	Recommendations change when interests or mobility change
4. Worker workflow	Case review, correction, referral and follow-up	Worker can turn a suggestion into a tracked action
5. Pilot hardening	Test noise, accents, interruptions, low connectivity and accessibility	Errors lead to repetition or human help, not fabricated answers

For the pitch, demonstrate two beneficiaries in the *same district* with different interests and travel constraints receiving different recommendations. Then show a worker correcting one misunderstood answer and the recommendation updating. Finish with the district view showing interest in a trade but no verified local batch: that directly illustrates the planning gap the project is designed to expose.

Success measures: interview completion rate, answer-correction rate, percentage of recommendations backed by verified qualifications, percentage with confirmed local availability, referral-to-enrolment rate, completion rate and subsequent livelihood outcomes. Label all demonstration records as synthetic; a functional prototype is not an official PM-AJAY integration or a guarantee of admission, funding or employment.

Screen layouts:





Pathway comparison and detail screens

When a beneficiary hesitates between multiple recommendations, this side-by-side view contrasts the commitments and practical realities without technical jargon.

Screen 7 — Pathway comparison matrix

text

← Back to recommendations [🔊 Read all]		
Comparing 2 selected options:		
Feature	Option A	Option B
Trade Name	Solar Tech	Food Prep
NSQF Level	Level 3	Level 2
Duration	320 Hours	210 Hours
Work Pattern	Wage / Company	Self-emp.
Travel Need	12 km (Bus)	Local/Home
Physical Need	Medium-High	Light

Batch Status

Starting Nov

Pending

[ Explain trade-offs in audio]

[Select Solar] [Select Food Prep]

- Audio breakdown: a single tap reads out a conversational trade-off summary (for instance: "Solar requires travel and offers steady contract wages, whereas Food Preparation can be operated from home with lower initial overhead").
- Visual anchors: large pictograms represent physical intensity, travel distance, and equipment requirements.

Screen 8 — Training centre and batch details

text

← Back

[ Hear info]

Rural Self-Employment Training Inst. (RSETI)

Sector 4, Industrial Area, District Centre

Distance: 8.5 km from your village

Transport: Direct bus route #14 (40 mins)

Batch Information:

- Course: Assistant Electrician (NSQF L3)
- Target Batch: Starting 15th November
- Intake: 25 seats (11 reserved for SC)
- Verification: Verified by District Team

Amenities available:

[✓ Free Tool Kit] [✓ Stipend] [✓ Hostel]

[ Request Call from Centre]

[ Send Location to WhatsApp]

- Verification badge: clearly indicates the timestamp when a field worker last verified instructor availability and hostel safety.
- Logistics strip: highlights daily transit feasibility and government-funded boarding entitlements under PM-AJAY Grant-in-Aid guidelines.

Post-referral beneficiary tracking

Beneficiaries often abandon training when administrative updates are silent; this screen provides an audio-enabled visual timeline of their progress.

Screen 9 — Beneficiary progress tracker

text

Application ID: PMAJAY-2026-8812	
Name: Sunita Devi Block: Moradabad Rural	
Your Journey Status:	
[✓]	1. Voice Counseling Complete Course selected: Tailoring & Apparel
[✓]	2. Field-Worker Verified Verified by Ramesh Kumar (VLE)
[●]	3. Batch Enrolment (Current Step) Documents submitted; awaiting start date
[○]	4. Skilling & Assessment 300-hour classroom & practical training
[○]	5. Tool Kit & Enterprise Linkage Seed support via GIA enterprise window
[📞]	Listen to latest update in Hindi
[💬]	Help: Contact My Village Coordinator

- Milestone indicators: high-contrast color nodes mark completed, active, and upcoming stages.
- Support hook: one-tap button triggers an outbound voice call from their assigned village-level worker if the application stalls.

Alternative channels and assisted workflows

To overcome limited smartphone ownership, these layouts support automated messaging channels and field-staff administration.

Screen 10 — WhatsApp voice note interface

text

WhatsApp: PM-AJAY Livelihood Bot (Verified) |

[Bot 10:14 AM] |

▶ ● 0:28 [Audio in Awadhi] |

"नमस्ते! आपकी रुचियों के आधार पर हमने 2 |

प्रशिक्षण कार्यक्रम चुने हैं..." |

[Beneficiary 10:15 AM] |

▶ ● 0:14 [Voice Note] |

"हमको गांव के पास सिलाई का काम सीखना है..." |

[Bot 10:16 AM] |

Quick options: |

[1. View Sewing Centres Near Me] |

[2. Connect with Moradabad Worker] |

[3. Hear Explanation in Hindi] |

 Attached: Livelihood-Summary-Card.pdf |

- Asynchronous voice processing: allows users to send voice notes in regional dialects, which the backend transcribes and answers with generated spoken notes.
- Action pills: simple number-coded reply chips eliminate text input requirements entirely.

Screen 11 — Field worker case review and intervention

text

Case ID: #7821 | Beneficiary: Rajesh Kumar |

Category: SC | Village: Chhajlet |

AI Audio Transcript (Confidence: 89%): |

"Heard: 8th pass, tractor repair experience, |

cannot travel more than 5 km daily." |

Field Corrections: |

Education: [Class 8 Pass ] |

Mobility: [Max 5 km ] |

Aspiration:[Agri-Mechanic / Ent ] |

Eligibility & Document Check:

☒ SC Caste Certificate / Self-Declaration

☒ Residence / Aadhaar Proof

☐ Income Criteria Verified

Assign Action:

☐ Select Training Partner / Centre

☐ Approve Referral

☐ Flag Ineligible

- Discrepancy flag: highlights low-confidence transcriptions so the worker can correct age, education, or travel radius before submission.
- Workflow dispatch: submits clean, verified beneficiary records straight into the district GIA training partner roster.

District administration and planning

This desktop view solves the core GIA planning dilemma by comparing grassroot beneficiary aspirations directly against available training seats.

Screen 12 — District GIA perspective planning dashboard

text

PM-AJAY District Planning Console | District: Moradabad | FY 2026-27

Beneficiaries Interviewed: 1,840 | Verified Matches: 1,410 | Gaps: 430

Trade Aspiration vs Training Supply:

Trade Name	Voice Demand	Sanctioned Seats	Supply Gap Status
Solar Technician	420	120	-300 (Severe Deficit)
Garment Making	510	480	-30 (Balanced)
Micro-Irrigation	290	0	-290 (No Centre)
Two-Wheeler Repair	380	200	-180 (Waitlisted)

Geographic Cluster Alerts:

• Block Bahjoi: 180 SC youth requested Agro-processing; nearest centre 45km

-> Recommendation: Deploy Mobile Training Unit under GIA Component

[Export Perspective Plan (CSV/PDF)] [Sanction New Batch Allocation]

- Supply gap alerts: explicitly surfaces instances where beneficiaries want a trade but no local GIA-funded batch exists.
- Data-driven sanctions: empowers administrators to allocate mobile training units or sanction additional batches where demand is demonstrated on the ground.